REWT

Compact memory and learned state refinement



Can energy refinement make compact memory more useful?

Kenju Tomita · Rochester Institute of Technology

The goal: remember a fact through distractions

A narrow memory task makes success and failure easy to inspect.

```
K=8;   X=5; X=9; ... many distractors ...   K=?
```

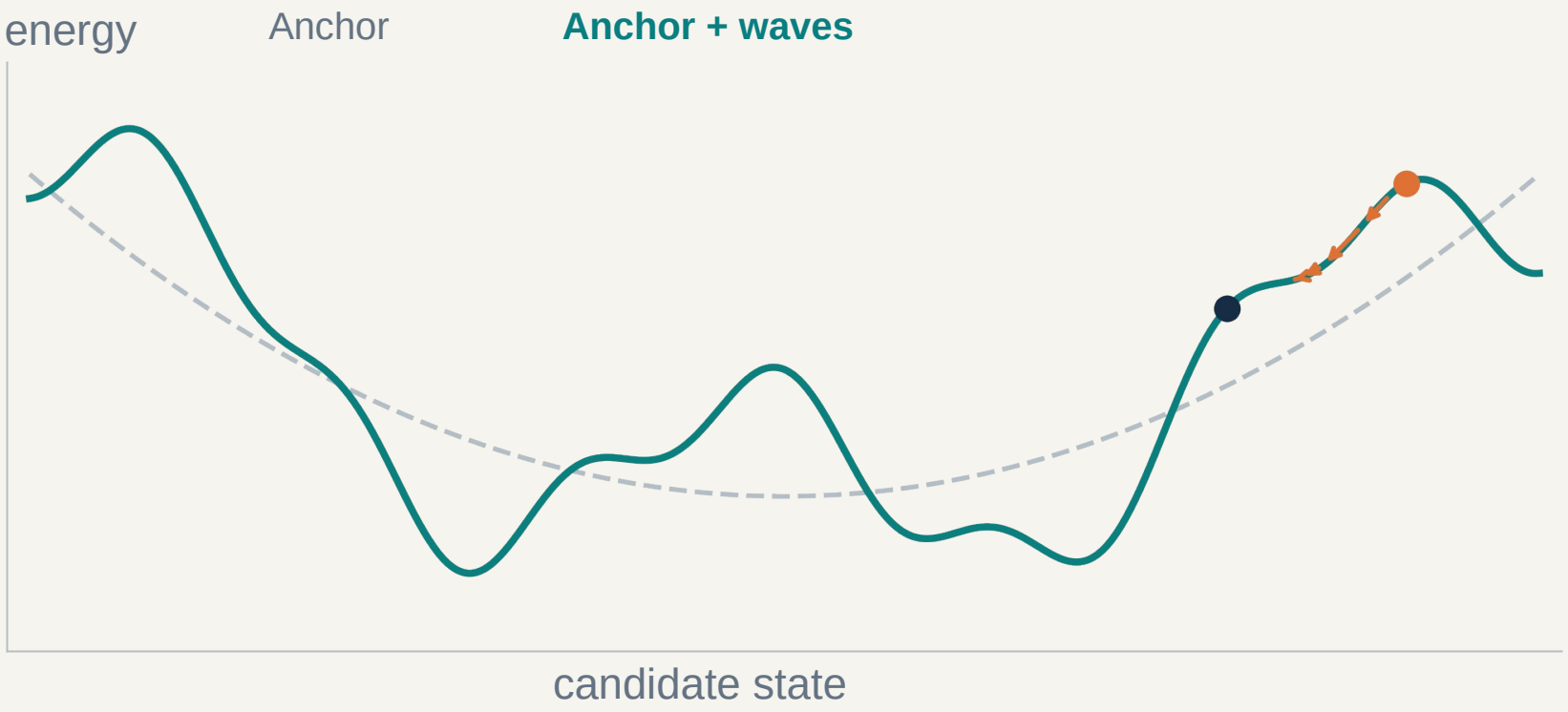


A compact memory should retain K without keeping every token.

Illustrative prompt. Recorded 64-distractor example: both hybrids return 8; full/window controls return 2/9.

Energy is a learned compatibility score

Lower energy means the model prefers that state. It does not certify truth.



THE LOOP

Guess → revise

Follow the slope.

Control the step size.

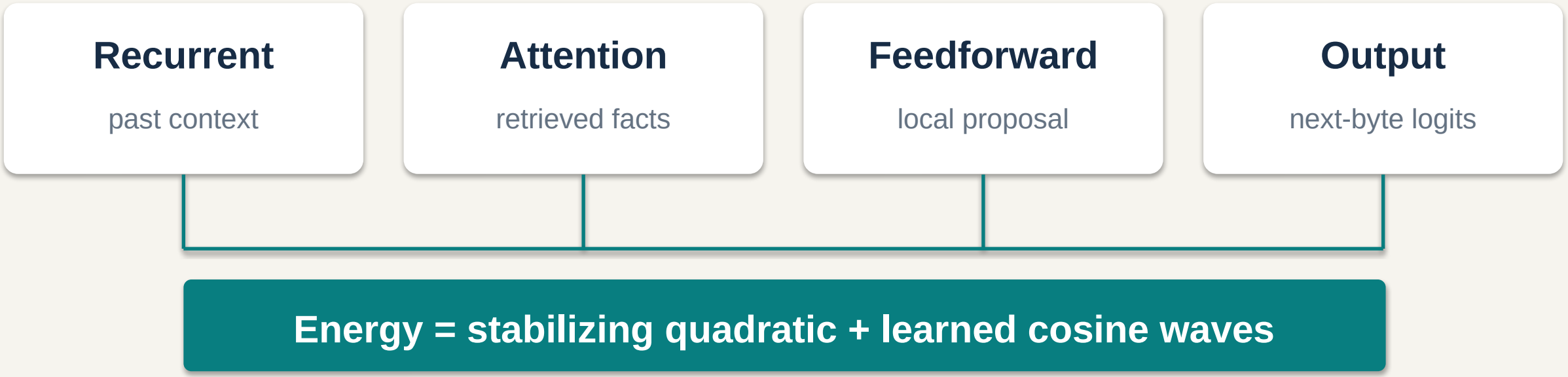
Keep weights fixed.

A lower local valley can still correspond to a wrong answer.

Illustrative landscape; the model state is high-dimensional. Background: Hoover et al., 2023.

The original model couples a joint workspace

One scalar energy scores recurrent, attention, feedforward and output states.



Current streaming models instead use a conventional backbone plus a final latent-energy head.

These are distinct architectures. The 1.784B conversion is not a fully joint energy transformer.

Training shapes the landscape. Inference moves the state.

TRAINING

Change parameters θ

Use labels or donor logits.
Differentiate through refinement.
Update trainable weights.

INFERENCE

Change candidate state s

Read the supplied context.
Take controlled state updates.
Keep learned weights fixed.

$$s \leftarrow s - \eta \nabla E\theta(\text{context}, s)$$

State optimization and parameter optimization are separate. Fixed-weight and analytic-gradient checks are automated.

Three questions need three different measurements

01 · NUMERICS

Does energy fall?

Check state updates and analytic gradients.

02 · QUALITY

Is the answer right?

Score generated answers, recall and format separately.

03 · COST

What does it cost?

Measure training and inference;

keep compute budgets visible.

Small models: three seeds | LFM3 recovery: one exploratory trajectory

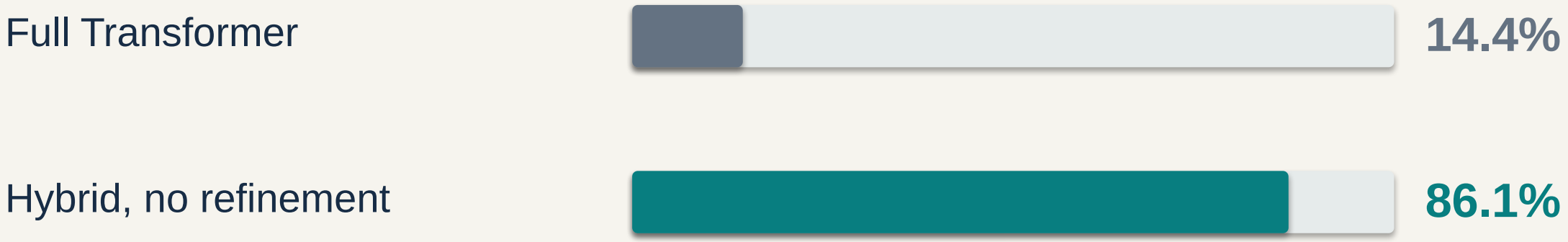
Report task scope, data splits, checkpoint selection and timing scope. Internal energy is not answer accuracy.

Streaming result · ~112K parameters

Compact memory helps on synthetic long recall

72 prompts with 32–128 distractors; averages across three training seeds.

K=8 … distractors … query K → 8



Hybrid + energy: 77.3% **Window Transformer: 21.3%**

Oct. 26: matched parameters, one common tuning recipe; weak third baseline seed. Recall tested to 520 input bytes, not 8K.

Execution result · same small-model weights

Faster decoding without retraining the model

Energy hybrid: generate 32 tokens after an 8K prefix, with two refinement rounds.

Warmed generation time	Previous	Optimized
CPU	15.43 ms	8.18 ms
GB10 GPU	24.51 ms	1.97 ms

About 225× less carried state than full attention at 8K.

Warmed decode only; excludes prefill/setup. Shared device. State excludes weights/workspace. Not a 1.7B model speed result.

Transfer pretrained weights into a streaming model

Standalone 1.784B-parameter REWT; no donor call during inference.

01 · INHERIT

LFM3-1.7B weights

tokenizer, embeddings, FFNs

02 · REPLACE

Delta + local attention

13 delta / 11 bounded attention

03 · RECOVER

Distill and adapt

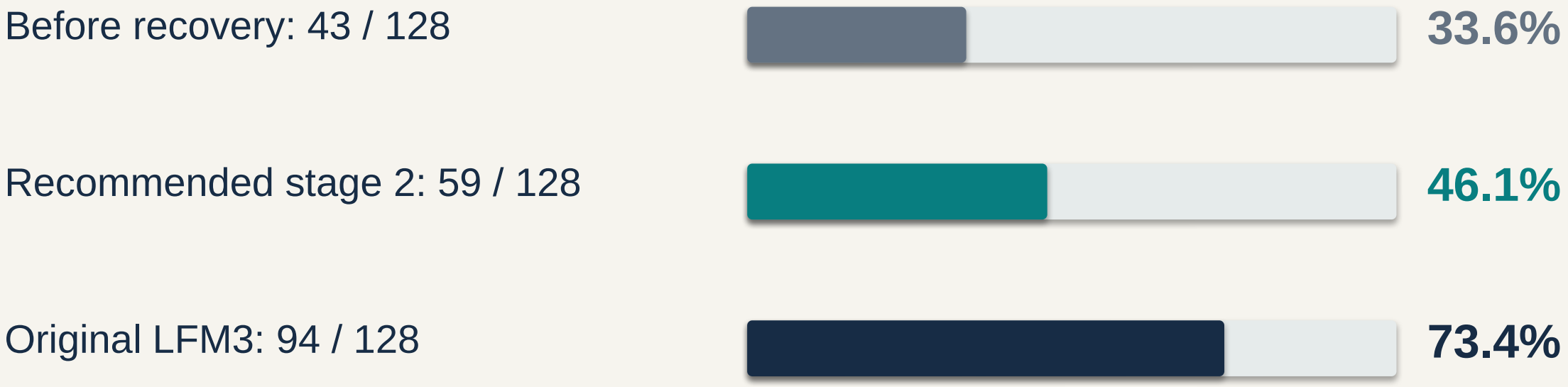
match logits; tune selected FFNs

256-token windows + compact delta memory + two energy updates

Compatible attention weights are inherited. Recovery changes weights; the architecture stays fixed across the recovery passes.

Recovery helps, but the donor gap remains

128 science questions, custom label ranking; same prompts and tokenizer for all models.



Exact-response probes: 9/50 → 13/50; original LFM3: 25/50.

Stage 2 includes 1.29M added token exposures. Recommendation used this small suite post hoc; no independent confirmation.

SCOPE

Partial recovery has a clear boundary

DEMONSTRATED

Measured progress

Compact synthetic recall.
Faster small-model decoding.
Partial pretrained-model recovery.

STILL OPEN

Reliable capabilities

Long recall in the converted model.
Tools, formatting and general reasoning.
An accuracy benefit from energy.

279 REWT tests pass. Implementation checks do not establish model quality.

Converted stage 2: 0/8 longer-recall codes and 0/4 tool-JSON probes. Exact-score gains mainly reflect short recall formatting.

TAKEAWAY

Make refinement earn its computational cost

1	**Teach correction**	Train energy to repair wrong candidates
2	**Recover memory**	Train varied long recall and structured outputs
3	**Compare fairly**	Fresh tests, zero-energy and equal-cost controls

Compact memory works here; energy must still earn its cost.

Research question: can energy refinement improve retrieval enough to justify its additional computation?

Lower energy is an internal guarantee

Hold the anchor and coefficients fixed during each refinement loop.

$$E(\delta) = \tfrac{1}{2}\,\delta^{T} P\,\delta + \Sigma\, a_k \cos(\omega_k^{T}\delta + \varphi_k)$$

δ = candidate − anchor P is positive diagonal

$$\delta' = \delta - \eta\,\nabla E(\delta), \quad \eta = 0.95 / L_{bound}$$

A bound on curvature controls the step. A lower score need not mean a better answer.

Streaming head: $L_{bound} = \max \text{diagonal}(P) + \Sigma\,|a_k|\,\|\omega_k\|^2$. This is distinct from the earlier joint-energy solver.

Energy does not yet improve the main task score

Small streaming models; all values average three seeds. Lower story loss is better.

Model	Long recall	Story bpb, equal CPU
Full Transformer	**14.35%**	**1.4641**
Hybrid, no refinement	**86.11%**	**1.4828**
Hybrid + energy	**77.31%**	**1.4926**

Frozen energy model: task accuracy is 36.63% with or without refinement.

Separate models measure training differences; disabling energy in one checkpoint measures inference contribution. Both matter.

Ideas are credited; project results are local

Hoover et al. · Energy Transformer · 2023

Yang et al. · Gated Delta Networks · ICLR 2025

Liquid AI · LFM3-1.7B · pinned donor checkpoint

Project evidence

Streaming comparison and decoder timings · October 26–27

LFM3 conversion, recovery and raw outputs · October 28

GitHub: kenjugmail/EnergyTransformer · reports/rewt_streaming_20261026 and reports/lfm3_capability_recovery_20261028